

Practical -12:

Out put: producer_consumer.c

```
akshitha_bushetty@LAPTOP-0FF1SH0T:~$ cd OSLab
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gedit practical12.c

** (gedit:621): WARNING **: 05:01:46.286: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found

** (gedit:621): WARNING **: 05:01:46.286: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gcc practical12.c -o practical12
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ ./practical12
PRODUCER-CONSUMER USING SEMAPHORES
=====
Buffer Size : 2
Items       : 10000
Time        : 0.1347 seconds
Throughput  : 74257.79 items/second
-----
Buffer Size : 5
Items       : 10000
Time        : 0.0402 seconds
Throughput  : 248613.98 items/second
-----
Buffer Size : 10
Items       : 10000
Time        : 0.0193 seconds
Throughput  : 517544.77 items/second
-----
Buffer Size : 20
Items       : 10000
Time        : 0.0072 seconds
Throughput  : 1380643.38 items/second
-----
```

deadlock.c

```
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gedit practical12b.c

** (gedit:1386): WARNING **: 05:03:52.641: Could not load Peas repository: T

** (gedit:1386): WARNING **: 05:03:52.641: Could not load PeasGtk repository
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gcc practical12b.c -o practical12b
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ ./practical12b
DEADLOCK DEMONSTRATION
=====
Thread 1: Trying to lock Resource A
Thread 1: Locked Resource A
Thread 2: Trying to lock Resource B
Thread 2: Locked Resource B
Thread 1: Trying to lock Resource B
Thread 2: Trying to lock Resource A
^C
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ |
```

deadlock_prevention

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akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gedit practical12.c
** (gedit:1823): WARNING **: 05:06:50.069: Could not load Peas re
** (gedit:1823): WARNING **: 05:06:50.069: Could not load PeasGtk
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ gcc practical12.c -o
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ ./practical12.c
DEADLOCK PREVENTION USING RESOURCE ORDERING
=====

Thread 1: Locking Resource A
Thread 1: Locked Resource A
Thread 2: Locking Resource A
Thread 1: Locking Resource B
Thread 1: Locked Resource B
Thread 1: Using both resources
Thread 1: Released resources
Thread 2: Locked Resource A
Thread 2: Locking Resource B
Thread 2: Locked Resource B
Thread 2: Using both resources
Thread 2: Released resources

No deadlock occurred.
akshitha_bushetty@LAPTOP-0FF1SH0T:~/OSLab$ |
```